PRINCE TAWIAH

Data Science | Analytics | Operations and Systems Reliability
Houghton, MI | (906) 319-8851 | ptawia3@mtu.edu | ptawiah.st@gmail.com
linkedin.com/in/ptawia3 | github.com/tbless19

PROFESSIONAL SUMMARY

M.S. Data Science candidate (Michigan Tech, Dec 2026; GPA 3.66) with an engineering background and 5+ years of experience turning large operational datasets into decisions. Builds predictive models, dashboards, and data pipelines in Python, SQL, and Spark, backed by hands-on IT and critical-infrastructure experience: hardware troubleshooting, Jira ticketing, VMware virtual machine clusters, fiber and power cabling, and uninterruptible power supply (UPS) and generator systems. Brings leadership of a 10-person technical team and the judgment to turn analysis into decisions on the ground.

EDUCATION

M.S. Data Science, Michigan Technological University **Expected Dec 2026**
GPA: 3.66 | Coursework: Machine Learning, Predictive Modeling, Advanced Data Mining, Statistical Programming and Analysis, Big Data Analytics, Computer Security

B.Sc. Telecommunication Engineering, Kwame Nkrumah University of Science and Technology **Jun 2018**

EXPERIENCE

Research Assistant, Michigan Technological University **Feb 2026 – Present**
- Design analytical frameworks to estimate IT power demand in large-scale data centers, modeling AI workload energy use across training, fine-tuning, and inference
- Develop time-series simulations of compute power from GPU and CPU utilization data, and build Python/HTML tools to visualize scenario results

Maintenance Technician, Zipline International Inc. **Dec 2019 – Jan 2025**
- Queried large-scale flight and logistics data with Snowflake SQL and analyzed flight metadata logs in Databricks to evaluate performance and surface inefficiencies
- Built dashboards used across 7 countries (10,000+ views) and key performance indicator (KPI) tracking for maintenance and inventory (6,000+ views), informing leadership decisions
- Supervised a team of 10 flight technicians as Maintenance Lead, sustaining >98% monthly fleet uptime with 0% maintenance-related aircraft retirements
- Logged and prioritized hardware issues in Jira and maintained site power (panels, UPS, generators) under strict lockout/tagout (LOTO) safety procedures

Interim Operations Manager, Zipline International Inc. **Jan 2022 – Dec 2024**
- Managed a team of 10 and led site operations during leadership gaps, using KPI data to prioritize work and sustain delivery targets

Quartermaster, Zipline International Inc. **Feb 2023 – Mar 2024**
- Achieved 100% inventory accuracy and zero stockouts, driving $70K+ in repair cost savings through disciplined tracking and return merchandise authorization (RMA) workflows

PROJECTS

Distributed CVD Risk Prediction using Apache Spark **PySpark, Scikit-learn**
- Deployed a 10-node Spark cluster and benchmarked 5 model pipelines (Logistic Regression, Random Forest, Gradient Boosting, SVM, Neural Network), reaching 85.06% accuracy and up to 5x faster processing

Galamsey Detection using CNN **Python, TensorFlow/Keras**
- Built a convolutional neural network (CNN) classifier on satellite imagery achieving 96% test accuracy and 0.99 ROC-AUC, validated with precision, recall, F1-score, and confusion matrices

School Website (k-ais.org), Airport International School, Kumasi **HTML, CSS, JavaScript, AWS**
- Designed, built, and deployed a public website for a real institution on AWS, including a careers page; maintain source in GitHub

SKILLS

Programming and Analytics: Python, SQL, R, Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PySpark, Matplotlib
IT and Infrastructure: VMware vCenter, Linux, Shell/Bash, Jira ticketing, hardware diagnostics, fiber and power cabling, fiber testing with an optical time-domain reflectometer (OTDR), UPS/generator systems, AWS (static site hosting)
Data Platforms and BI: Snowflake, BigQuery, Databricks, Apache Spark, Tableau, Power BI, Looker, Excel, Git

AWARDS

Best Flight Operator, Zipline Ghana (2023) | Customer Obsession and Servant Leadership Award, Zipline Ghana (2022)